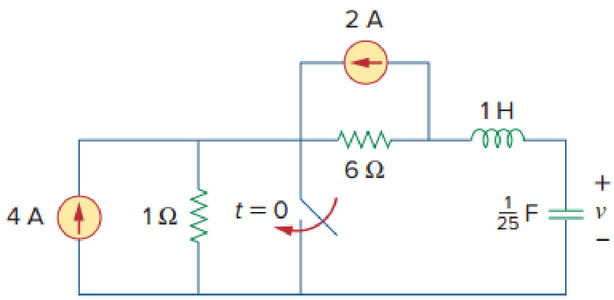
Problem

Given the network in Fig. 8.90, find $v(t)$ for $t > 0$.

Figure 8.90:



Step-by-step solution

Step 1 of 13

Draw the following circuit diagram:

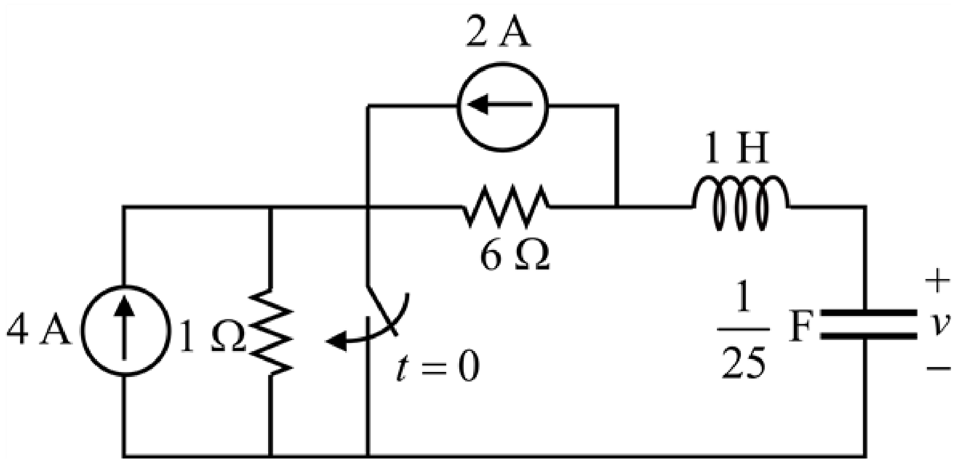


Figure 1

Step 2 of 13

Use the Source transformation technique and redraw the Figure 1 as shown in Figure 2.

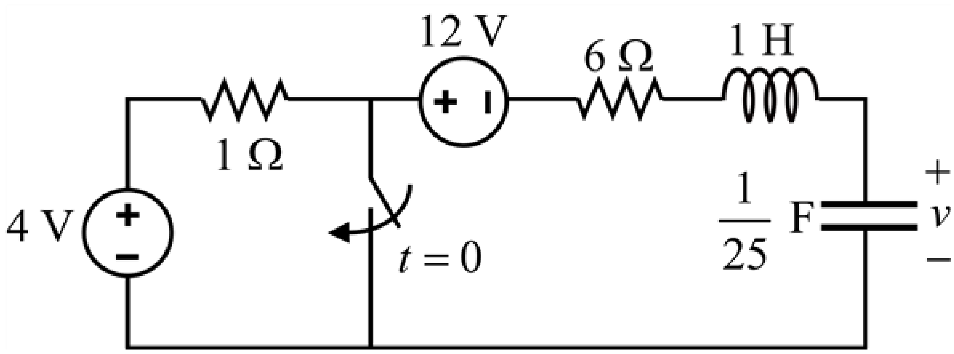


Figure 2

Step 3 of 13

For $t<0$, the switch in Figure 2 is open for a long time and the circuit is in Steady state.

The capacitor behaves like an open circuit and the inductor acts like a short circuit.

Redraw the Figure 2 as shown in Figure 3.

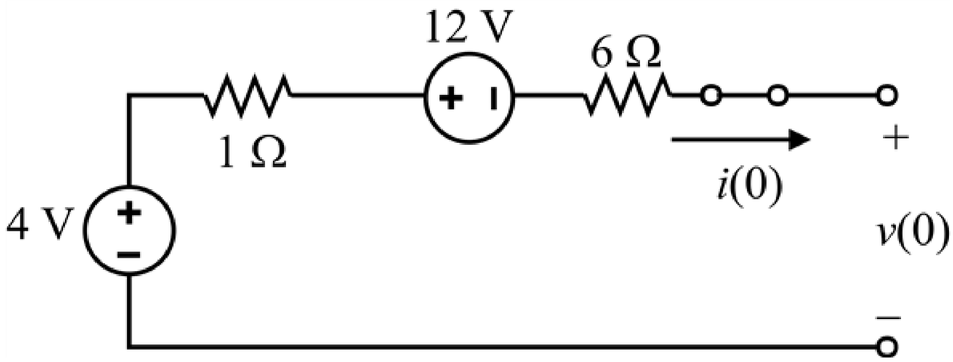


Figure 3

Step 4 of 13

Calculate the value of the initial current through the inductor $i(0)$.

Clearly from Figure 3, $i(0)=0\text{ A}$.

Calculate the value of the initial voltage across the capacitor $v(0)$.

Apply Kirchhoff Voltage Law to Figure 3.

$$-4\text{ V}+12\text{ V}+v(0)=0$$
$$v(0)=-8\text{ V}$$

Therefore, the value of the initial voltage across the capacitor $v(0)$ is -8 V.

Step 5 of 13

For $t>0$, the switch in Figure 2 is closed and redraw the Figure 2 as shown in Figure 4.

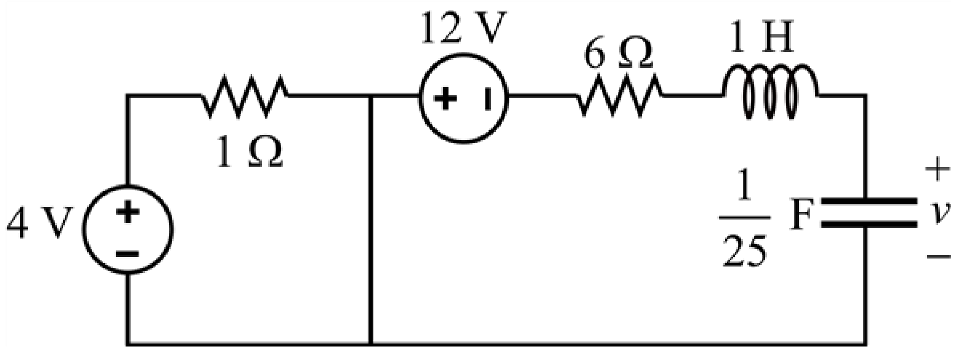


Figure 4

Step 6 of 13

Draw the modified diagram of Figure 4 as shown in Figure 5.

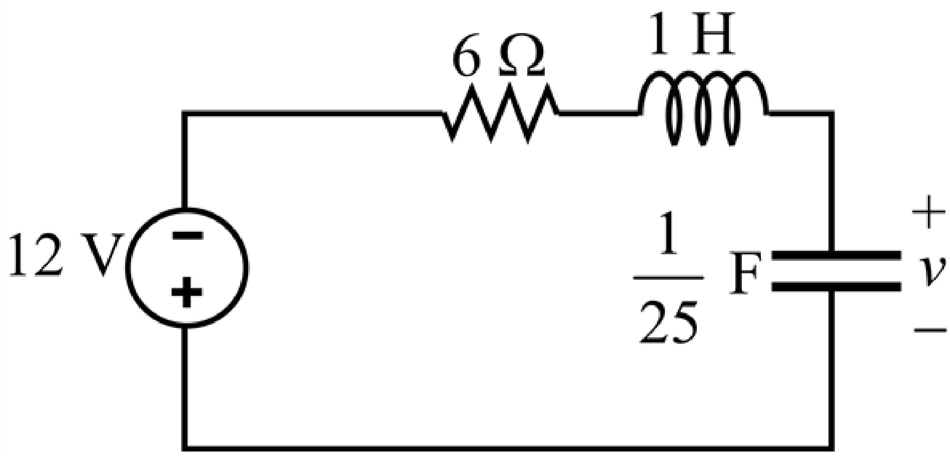


Figure 5: A series RLC circuit.

Clearly from Figure 5, a series RLC circuit is remained with a voltage source of -12 V.

Calculate the value of the damping factor α.

Write the expression for the damping factor α.

$$\alpha = \frac{R}{2L}$$

Substitute $6\,\Omega$ for R, 1 H for L in this expression.

$$\begin{aligned}\alpha &= \frac{6}{2(1)} \\ &= 3\end{aligned}$$

Therefore, the value of the damping factor α is 3 Np/s.

Step 7 of 13

Calculate the value of the undamped natural frequency ω_0.

Write the expression for the undamped natural frequency ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 1 H for L, $\frac{1}{25}\text{ F}$ for C in this expression.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{1\left(\frac{1}{25}\right)}} \\ &= \sqrt{25} \\ &= 5\text{ rad/s}\end{aligned}$$

Therefore, the value of the undamped natural frequency ω_0 is 5 rad/s.

Step 8 of 13

The value of damping factor α is 3 Np/s and it is less than value of the undamped natural frequency $\omega_0 = 5\text{ rad/s}$.

Hence, the response is underdamped $v(t)$.

Therefore, Write the expression for the underdamped response $v(t)$.

$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$ ...... (1)

Where,

V_s is the final value of $v(t)$,

ω_d is damping frequency,

α is damping factor.

Step 9 of 13

Calculate the final value of $v(t)$.

The final value of $v(t)$ is same as the source voltage V_s.

Hence, $V_s = -12\text{ V}$.

Step 10 of 13

Calculate the value of the damping frequency ω_d.

Write the expression for the damping frequency ω_d.

$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

Substitute 5 rad/s for ω_0, 3 Np/s for α in this expression.

$$\begin{aligned}\omega_d &= \sqrt{5^2 - 3^2}\\ &= \sqrt{16}\\ &= 4\end{aligned}$$

Therefore, the value of the damping frequency ω_d is 4 rad/s.

Step 11 of 13

Substitute -12 V for V_s, 4 rad/s for ω_d, 3 Np/s for α in equation 1.

$v(t) = -12 + (A_1 \cos 4t + A_2 \sin 4t)e^{-3t}$

Substitute 0 for t in this expression.

$$\begin{aligned}v(0) &= -12 + (A_1 \cos 4(0) + A_2 \sin 4(0))e^{-3(0)}\\ &= -12 + (A_1(1) + A_2(0))(1)\\ &= -12 + A_1\end{aligned}$$

Substitute -8 V for $v(0)$ in this expression.

$$\begin{aligned}-8 &= -12 + A_1\\ A_1 &= 4\end{aligned}$$

Therefore, the value of A_1 is 4.

Step 12 of 13

Calculate the value of the $\frac{dv(0)}{dt}$.

Differentiate $v(t)$ with respect to t to find $\frac{dv(t)}{dt}$.

$$\begin{aligned}\frac{dv(t)}{dt} &= \frac{d\left(-12+\left(A_1\cos 4t + A_2\sin 4t\right)e^{-3t}\right)}{dt} \\ &= -3\left(A_1\cos 4t + A_2\sin 4t\right)e^{-3t} + \left(-4A_1\sin 4t + 4A_2\cos 4t\right)e^{-3t} \\ &= \left[\left(-3A_1 + 4A_2\right)\cos 4t + \left(-4A_1 - 3A_2\right)\sin 4t\right]e^{-3t}\end{aligned}$$

Substitute 0 for t in this expression.

$$\begin{aligned}\frac{dv(0)}{dt} &= \left[\left(-3A_1 + 4A_2\right)\cos 4(0) + \left(-4A_1 - 3A_2\right)\sin 4(0)\right]e^{-3(0)} \\ &= \left[\left(-3A_1 + 4A_2\right)(1) + \left(-4A_1 - 3A_2\right)(0)\right](1) \\ &= -3A_1 + 4A_2\end{aligned}$$

Substitute 4 for A_1, $\frac{i(0)}{C}$ for $\frac{dv(0)}{dt}$ in the expression: $\frac{dv(0)}{dt} = -3A_1 + 4A_2$

$$\frac{i(0)}{C} = -3(4) + 4A_2$$

Substitute 0 for $i(0)$ in this expression.

$$\begin{aligned}\frac{(0)}{C} &= -12 + 4A_2 \\ -12 + 4A_2 &= 0 \\ 4A_2 &= 12 \\ A_2 &= 3\end{aligned}$$

Therefore, the value of A_2 is 3.

Step 13 of 13

Substitute 4 for A_1, 3 for A_2 in the expression: $v(t) = -12 + \left(A_1\cos 4t + A_2\sin 4t\right)e^{-3t}$

$$v(t) = -12 + \left(4\cos 4t + 3\sin 4t\right)e^{-3t}$$

Therefore, the required value of the response $v(t)$ is $\boxed{-12 + \left(4\cos 4t + 3\sin 4t\right)e^{-3t}\ \text{V}}$.